IBA COLLEGE OF MINDANAO INC.
College of Information Technology

BIOMETRIC ATTENDANCE SYSTEM WITH VISUAL STUDIO INTEGRATION

Subject: Systems Integration and Architecture

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July 202

Part I – Organization and Project Background

Project Title

Biometric Attendance System with Visual Studio Integration

Organization Name

College of Information Technology (CIT), IBA College of Mindanao, Inc.

Nature of the Organization

College of Information Technology (CIT) at IBA College of Mindanao, Inc. It offers undergraduate education in Information Technology, equipping students with knowledge and practical skills in programming, networking, database management, systems analysis, cybersecurity, and software development. The program is supported by faculty members, laboratory personnel, and administrators who manage student records, attendance, academic performance, and laboratory activities to ensure quality education and efficient academic operations.

Existing Business Process

Currently, attendance in the BSIT program is recorded manually by instructors at the beginning of each class. Students sign an attendance sheet or answer a roll call, after which the instructor verifies and records their attendance. At the end of the day or semester, attendance records are manually encoded into a computer and compiled to generate attendance reports. Faculty members use these reports to monitor student attendance and determine compliance with school policies. This process requires significant time and effort and relies heavily on manual record-keeping.

Current Problems

The current attendance system of the BSIT program relies on manual recording, which takes time and may cause errors in attendance data. It is also vulnerable to proxy attendance, delayed record updates, and difficulties in generating and retrieving attendance reports. Paper-based records may also be lost or damaged, affecting data accuracy and reliability.

Objectives of the Proposed IoT Solution

The **Biometric Attendance System with Visual Studio Integration** aims to automate attendance recording using fingerprint authentication and IoT technology. The system will provide real-time attendance monitoring, secure data storage, automatic report generation, and a Visual Studio-based application to improve accuracy, security, and efficiency while reducing the workload of instructors.

Part II – Stakeholder Analysis

Stakeholder	Role	Responsibilities	Expectations
Management (BSIT Department)	Oversees the implementation of the system	Approves the system project and monitors its effectiveness in improving attendance management.	Expect accurate attendance monitoring and improved academic management.
System Administrator	Manages the attendance system and database	Maintains user accounts, manages records, and ensures the system operates properly.	Expect a secure, reliable, and easy-to-maintain system.
Network Administrator	Manages network connectivity	Ensures stable communication between IoT devices, computers, and servers.	Expect secure and reliable network performance.
IoT Administrator	Manages IoT devices and connections	Configures, monitors, and maintains biometric devices and IoT communication.	Expect proper device operation and accurate data transmission.
End Users (Students)	Main users of the system	Register biometric information and use fingerprint scanning to record attendance.	Expect a fast, accurate, and convenient attendance process.
Customers (Students/School Community)	Beneficiaries of the system	Use the improved attendance service provided by the institution.	Expect better attendance tracking and reliable records.
Government Agencies (if applicable)	Regulatory stakeholders	Ensure compliance with education and data privacy regulations.	Expect the system to follow legal and security requirements.
Third-party Cloud Providers	Provides cloud services for data storage and backup (if used)	Maintains cloud infrastructure and ensures data availability.	Expect secure data management and reliable service availability.

Part III – Requirements Analysis (20 Points)

Functional Requirements

- The system shall collect fingerprint data from biometric sensors.
- The system shall register student information and biometric records.
- The system shall authenticate students using fingerprint verification.
- The system shall automatically record attendance after successful verification.
- The system shall transmit attendance data through IoT communication.
- The system shall store attendance records in a centralized database.
- The system shall display real-time attendance monitoring.
- The system shall generate attendance reports.
- The system shall provide user authentication for administrators.
- The system shall support API communication between devices and applications.
- The system shall allow searching and viewing previous attendance records.
- The system shall provide backup of attendance data.
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Non-Functional Requirements

- **Security** – The system must protect student information and biometric data.
- **Reliability** – The system must provide accurate and consistent attendance recording.
- **Availability** – The system should be accessible during class hours.
- **Performance** – The system should process fingerprint verification quickly.
- **Low Latency** – Attendance updates should appear with minimal delay.
- **Scalability** – The system should support increasing numbers of students.
- **Data Privacy** – Student biometric information must be handled securely.
- **Backup and Recovery** – The system should recover data after failures.
- **Fault Tolerance** – The system should continue working during minor errors.
- **Maintainability** – The system should be easy to update and troubleshoot.
- **Usability** – The system interface should be simple and user-friendly.
- **Compatibility** – The system should work with IoT devices, databases, and Visual Studio applications.